DATABASE SYSTEMS PROJECT

Health Information Systems

LABORATORY INFORMATION SYSTEM (LIS)

PART 1: Database Selection

1a). Area of Interest: Health Information Systems

The selected area of interest is Health Information Systems a domain encompassing the collection, storage, management, and transmission of health-related data in clinical and administrative settings. This domain is critical in modern healthcare, where timely and accurate data directly influences patient outcomes and organizational efficiency.

1b). Possible Database Systems in Health Information Systems

The following are some of the database systems applicable within the Health Information Systems domain:

#	System	Description
1	Electronic Medical Records Database	Manages patient histories, diagnoses, prescriptions and clinical notes
2	Laboratory Information System (LIS)	Handles test orders, specimen tracking, results, and lab reporting
3	Hospital Management Information System (HMIS)	Covers admissions, bed management, billing and hospital-wide operations
4	Pharmacy Information System (PIS)	Tracks drug inventory, prescriptions and medication dispensing

#	System	Description
5	Health Surveillance & Disease Reporting System	Aggregates disease incidence data for public health monitoring

1c). Selected system: Laboratory Information System (LIS):

A Laboratory Information System manages the complete workflow of a clinical or diagnostic laboratory from physician test requests through sample collection, analysis, result validation, and final report delivery. Reasons for this selection include:

- High data volume and complexity: Laboratories generate thousands of structured records daily (patients, tests, results,), making rigorous database design essential.
- Direct clinical impact: Laboratory results inform approximately 70% of clinical decisions, so a reliable LIS directly improves patient outcomes.
- Rich relational structure: The LIS domain has clearly defined entities with well-structured relationships, making it an ideal relational database case study.
- Local relevance: Many Ugandan health facilities still rely on paper-based laboratory systems; a digital LIS is both timely and impactful in this context.
- Demonstrable SQL queries: The LIS supports meaningful queries — pending results, abnormal findings, TAT reports, enabling comprehensive SQL demonstration.

PART 2: Business Case for the Laboratory Information System

2a. Application Domain

The LIS supports the operations of a clinical diagnostic laboratory within a hospital or standalone diagnostic centre. It automates the following tasks, operations, and processes:

- Test Order Management: Physicians submit test requests electronically. The LIS assigns accession numbers and initiates the workflow.
- Specimen Tracking: Each biological sample is tracked from collection through transportation, processing, and analysis using barcoded labels.
- Result Entry and Validation: Laboratory scientists enter or review results, with the system auto-flagging abnormal values against configurable reference ranges.

- Report Generation and Delivery: Finalized results are compiled into formatted reports delivered electronically or printed for the requesting clinician.
- Quality Control (QC) Management: QC runs are recorded and monitored; out-of-range failures prevent result release automatically.
- Inventory and Reagent Tracking: Reagent lot numbers, expiry dates, and usage quantities are tracked to ensure consumable availability.

User groups served:

User Group	Role in the System
Laboratory Scientists	Result entry, validation, QC management, instrument monitoring
Laboratory Technicians	Sample reception, specimen processing, test performance
Physicians / Clinicians	Test ordering, result viewing, clinical decision-making
Nurses / Ward Staff	Sample collection requests and patient identification
Laboratory Manager	Workload reporting, staff performance, equipment utilization, QC review
Hospital Administrator	Billing integration, laboratory revenue tracking, compliance reporting
Patients	Indirect beneficiaries — receive faster, more accurate diagnostic reports

2b. Analysis of Benefits

Tangible Benefits

- Reduced Turnaround Time (TAT): Automation reduces average TAT from 4–6 hours (manual) to 1–2 hours, enabling faster clinical decisions.
- Error Reduction: Eliminating manual transcription reduces clerical result errors by an estimated 60–80%, reducing costly repeat tests.

- Increased Throughput: Automated order management allows the same staff to process 30–50% more samples per shift.
- Improved Billing Accuracy: Every test ordered and performed is captured, reducing revenue leakage from unbilled tests.

Intangible Benefits

- Improved Patient Safety: Accurate, timely results reduce the risk of misdiagnosis and inappropriate treatment.
- Staff Satisfaction: Laboratory staff spend less time on paperwork and more time on analytical and interpretive work.
- Organisational Reputation: A technology-enabled laboratory attracts more referrals from external practitioners.
- Research Data: Aggregated, de-identified laboratory data supports epidemiological research and disease surveillance.

2c. Cost Analysis

Cost Category	Estimated Cost (UGX)	Type
System Development / Licensing	15,000,000 – 30,000,000	Tangible / One-time
Server Hardware & Infrastructure	8,000,000 – 20,000,000	Tangible / One-time
Network Setup (LAN/Wi-Fi)	3,000,000 – 8,000,000	Tangible / One-time
Staff Training	2,000,000 – 5,000,000	Tangible / One-time
Annual Maintenance & Support	3,000,000 – 5,000,000/yr	Tangible / Recurring
Workflow Disruption during Go-live	2–3 weeks reduced efficiency	Intangible

Cost Category	Estimated Cost (UGX)	Type
Staff Resistance / Learning Curve	Temporary productivity dip	Intangible

Cost-Benefit Justification: Estimated annual benefits (error reduction, billing recovery, throughput gains) are 25,000,000–60,000,000 UGX, comfortably exceeding annual running costs of 3,000,000–8,000,000 UGX. Capital costs are typically recovered within 18–30 months, confirming economic feasibility.

2d. User Issues

- Computer Literacy: Staff vary in digital skills. Training must be tiered — basic navigation for technicians, advanced administration for managers.
- Adoption Resistance: Staff accustomed to paper workflows may resist change. Early staff involvement in system design and identification of change champions are essential mitigation strategies.
- Workflow Redesign: The LIS replaces paper-based processes such as result phoning and logbook recording; these must be formally retired.
- Access Control: Strict role-based access controls must be enforced to protect patient data privacy and prevent unauthorised result modification.
- Parallel Running: Both paper and electronic systems should run simultaneously for 2–4 weeks during go-live to build confidence and catch discrepancies.

2e. Risk Assessment

Risk	Category	Mitigation Strategy
System downtime during critical lab hours	Technical	Redundant servers, UPS power supply, daily backups
Data loss or corruption	Technical	Automated nightly backups to offsite / cloud storage

Risk	Category	Mitigation Strategy
Staff refusal to adopt the system	Organizational	Change management programme, management endorsement, incentives
Budget overrun during implementation	Organizational	Phased implementation, fixed-price vendor contracts
Patient data breach / privacy violation	Legal / Ethical	Encryption, role-based access, audit logging, staff NDAs
Regulatory non-compliance	Legal	Build to Uganda MOH and ISO 15189 standards from the outset

2f. System Limitations

- Single-site Scope: The system is designed for one laboratory facility. Multi-site networked deployment requires additional architectural design.
- No Billing Module: Financial integration with hospital billing or insurance systems is not included in this design.
- No Patient Portal: Patients cannot directly access results online; clinician-mediated delivery is the only supported workflow.
- No AI / Clinical Decision Support: The system stores and retrieves data but does not perform predictive analytics or AI-based critical-value alerting.
- English Only: The system is not localised for Luganda or other Ugandan languages.

PART 3: Database Design — Logical Design

3. Entities and Their Attributes

PATIENT

Attribute	Data Type	Description
patient_id (PK)	INT AUTO_INCREMENT	Unique patient identifier

Attribute	Data Type	Description
patient_name	VARCHAR(50) NOT NULL	Patient's name
date_of_birth	DATE NOT NULL	Patient's date of birth
gender	ENUM('M','F','O')	Patient's gender
phone	VARCHAR(15)	Contact telephone number
address	TEXT	Residential address
national_id	VARCHAR(20) UNIQUE	National identification number
blood_group	VARCHAR(5)	ABO blood group (e.g. O+)

DOCTOR

Attribute	Data Type	Description
doctor_id (PK)	INT AUTO_INCREMENT	Unique doctor identifier
doctor_name	VARCHAR(50) NOT NULL	Doctor's name
phone	VARCHAR(15)	Contact number
department_id (FK)	INT	Hospital department the doctor belongs to

DEPARTMENT

Attribute	Data Type	Description
department_id (PK)	INT AUTO_INCREMENT	Unique department identifier

Attribute	Data Type	Description
department_name	VARCHAR(100) NOT NULL	Section name (e.g. Haematology)
description	TEXT	Functions of the department
head_staff_id (FK)	INT	Staff member who heads the department

LAB_TECH

Attribute	Data Type	Description
staff_id (PK)	INT AUTO_INCREMENT	Unique staff identifier
tech_name	VARCHAR(50) NOT NULL	Staff member's name
role	ENUM('Scientist','Technician','Manager')	Job role within the laboratory
department_id (FK)	INT	Section the staff member is assigned to

TEST

Attribute	Data Type	Description
test_id (PK)	INT AUTO_INCREMENT	Unique test identifier
test_name	VARCHAR(100) NOT NULL	Full name (e.g. Full Blood Count)
sample_type	VARCHAR(50)	Required specimen (e.g. Blood, Urine)
cost	DECIMAL(10,2)	Test fee in UGX

Attribute	Data Type	Description
department_id (FK)	INT	Laboratory section that performs the test

TEST_ORDER

Attribute	Data Type	Description
order_id (PK)	INT AUTO_INCREMENT	Unique order identifier
patient_id (FK)	INT NOT NULL	Patient being tested
doctor_id (FK)	INT NOT NULL	Requesting physician
order_date	DATETIME DEFAULT CURRENT_TIMESTAMP	Date and time order was placed
status	ENUM('Pending','In Progress','Completed','Cancelled')	Current workflow status
clinical_notes	TEXT	Clinical context or diagnosis hint from doctor

ORDER_TEST_ITEM (Bridge resolves M:N between TEST_ORDER and TEST)

Attribute	Data Type	Description
order_id (PK, FK)	INT NOT NULL	References TEST_ORDER
test_id (PK, FK)	INT NOT NULL	References TEST

SAMPLE

Attribute	Data Type	Description
sample_id (PK)	INT AUTO_INCREMENT	Unique sample identifier
accession_number	VARCHAR(20) UNIQUE	Lab-generated barcode / accession number
order_id (FK)	INT NOT NULL	Associated test order
sample_type	VARCHAR(50)	Type of specimen collected
collection_datetime	DATETIME	When sample was collected
collected_by (FK)	INT	Lab staff who collected the sample
condition	ENUM('Acceptable','Haemolysed','Insufficient','Mislabelled')	Sample quality on receipt
received_datetime	DATETIME	When sample was

Attribute	Data Type	Description
		received in the lab

RESULT

Attribute	Data Type	Description
result_id (PK)	INT AUTO_INCREMENT	Unique result identifier
sample_id (FK)	INT NOT NULL	Sample on which test was performed
test_id (FK)	INT NOT NULL	Test that was performed
result_value	VARCHAR(100)	Reported result (numeric or descriptive text)
flag	ENUM('Normal','High','Low','Critical')	Interpretation relative to reference range
result_datetime	DATETIME DEFAULT CURRENT_TIMESTAMP	When result was entered or generated
validated_by (FK)	INT	Lab scientist who validated the result
comments	TEXT	Additional interpretive comments

REPORT

Attribute	Data Type	Description
report_id (PK)	INT AUTO_INCREMENT	Unique report identifier
order_id (FK)	INT NOT NULL	Associated test order
generated_by (FK)	INT	Staff who generated the report

Attribute	Data Type	Description
generated_datetime	DATETIME DEFAULT CURRENT_TIMESTAMP	Date and time of report generation
delivery_method	ENUM('Print','Email','Portal')	How report was delivered
status	ENUM('Draft','Final','Amended')	Report finalization status

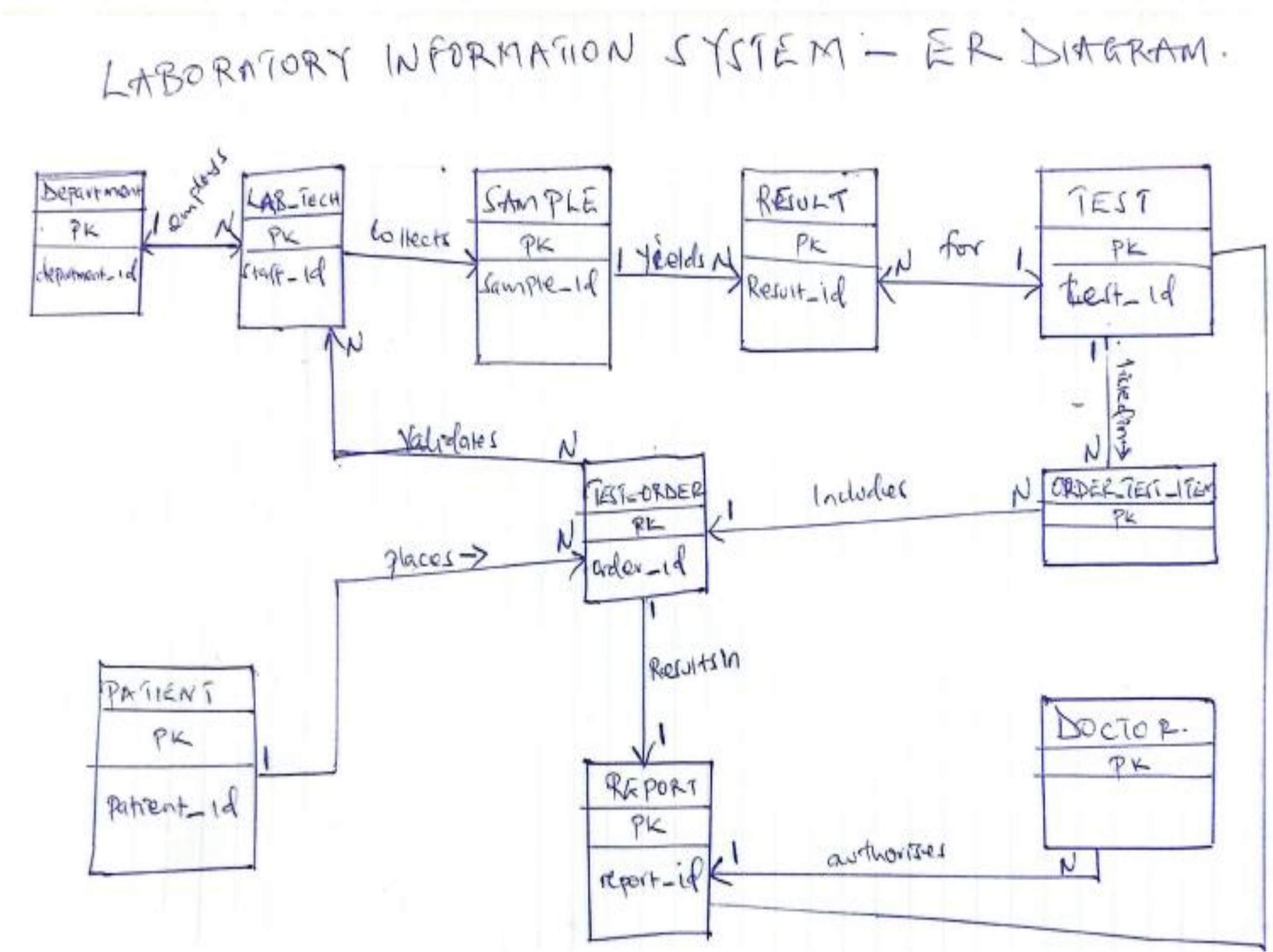
3c & 3d. Entity Relationships with Multiplicities

All applicable relationship types are identified below:

Entity A	Verb	Entity B	Type	Multiplicity
PATIENT	places	TEST_ORDER	Binary	1 : Many (1:N)
DOCTOR	authorises	TEST_ORDER	Binary	1 : Many (1:N)
TEST_ORDER	produces	SAMPLE	Binary	1 : Many (1:N)
TEST_ORDER	includes	TEST	Binary (M:N)	Many : Many via ORDER_TEST_ITEM
SAMPLE	yields	RESULT	Binary	1 : Many (1:N)
TEST	defines	RESULT	Binary	1 : Many (1:N)
LAB_TECH	collects	SAMPLE	Binary	1 : Many (1:N)
LAB_TECH	validates	RESULT	Binary	1 : Many (1:N)
DEPARTMENT	classifies	TEST	Binary	1 : Many (1:N)
DEPARTMENT	employs	LAB_TECH	Binary	1 : Many (1:N)
TEST_ORDER	results in	REPORT	Binary	1 : 1 (one-to-one)
LAB_TECH	generates	REPORT	Binary	1 : Many (1:N)

Entity A	Verb	Entity B	Type	Multiplicity
PATIENT + DOCTOR + TEST	constitutes	TEST_ORDER	Ternary	Many:Many:Many via TEST_ORDER
DEPARTMENT supervises itself	(head-of)	DEPARTMENT	Unary / Recursive	1 : Many (one head, many staff)

The ER DIAGRAM



3f. User Interface — Dialogue Structure

The LIS interface is organised around key laboratory workflows. The table below describes the main modules and their screens:

UI Module	Key Screens and Functions
Login / Authentication	Username + password; role-based access (Scientist, Technician, Manager, Admin)
Patient Registration	Search existing patient or register new; capture demographics, national ID, blood group
Test Order Entry	Select patient, doctor, test(s); set priority; add clinical notes; confirm order
Sample Reception	Scan accession barcode; verify sample condition; assign to department work queue
Result Entry	Enter results per test; auto-flag vs reference range; attach equipment used
Result Validation	Scientist reviews flagged/abnormal results; approves or sends for repeat testing
Report Generation	System compiles all results for an order; formats report; selects delivery method
Search / Query	Search by patient ID, accession number, date range, test type, or status
Manager Dashboard	Workload stats, TAT analysis, abnormal result rates, equipment utilisation
Administration	Manage users, reference ranges, test catalogue, departments, equipment records

3g. Database Schema — Physical Design

The physical schema is implemented in MySQL. The table below shows each table with its primary key and foreign key references:

Table	Primary Key	Foreign Keys
patient	patient_id	—
doctor	doctor_id	department_id → department
department	department_id	head_tech_id → lab_tech
lab_tech	tech_id	department_id → department
test	test_id	department_id → department
test_order	order_id	patient_id → patient; doctor_id → doctor
order_test_item	(order_id, test_id)	order_id → test_order; test_id → test
sample	sample_id	order_id → test_order; collected_by → lab_tech
result	result_id	sample_id → sample; test_id → test; validated_by → lab_tech;
report	report_id	order_id → test_order; generated_by → lab_tech

3h. Description of Queries and Reports

The five key queries below directly address user and organisational needs identified in the business case:

Query / Report	Business Need Addressed
Q1: All pending test orders (sorted by priority)	Workload management; ensures STAT and Urgent samples are actioned first (Section 2b)
Q2: Critical / abnormal results — last 7 days	Patient safety; gives clinicians and managers a critical-value tracking view (Section 2b)
Q3: Average turnaround time by test type	Quality management; compares actual TAT vs targets for ISO 15189 compliance (Section 2b)

Query / Report	Business Need Addressed
Q4: Monthly workload and revenue by department	Managerial reporting; justifies staffing budgets and tracks revenue recovery (Section 2b)